



[The exam text is in English. You should answer the questions in English.]

The exam consists of 13 questions (8 pages). Please first check if any page is missing in the printout you. For the open questions: please keep the handwriting clear and readable.

The exam starts at 9.00 hrs and ends at 12.00 hrs. Participation in the exam requires being present for **at least 1 hour**.

Electronic devices are not allowed and must be stowed.

Points for each open question are indicated.

- Please write readable and write your name and student identification number on every single sheet of paper for unique identification!
- **Don't spend much time writing long stories – that's NOT necessary. Don't panic! You CAN make it!**
- Look at the whole exam first to get an overview, pay attention to the grading scheme of multiple-choice questions, and do those assignments first which you feel are doable for you.
- Good success and all the best for your studies!

Part I - Multiple-choice [50 points].

- Each problem has a variable number of correct statements. Please write your answers in the table given at the end of the question sheet.
- In NO problem, all statements are correct. If you choose all the options, i.e., "ABCD", the score for that question will be ZERO!
- The score for each question will be given according to Morgan method:

$$5 \times \max \{ r / \#correct\ option(s) - w / \#distractors(s), 0 \},$$

where:

- 5 is the total score for each question;
- r is the number of option(s) you select correctly;
- w is the number of option(s) you select incorrectly;
- $\#correct\ option(s)$ is the number of correct options of this question;
- $\#distractors(s)$ is the number of incorrect options/distractors of this question;
- Please DO NOT mark an option if you are not certain with it as a penalty will incur.

Please write the alphabetic characters (e.g., CD) representing your answers of the multiple-choice questions into the table below.

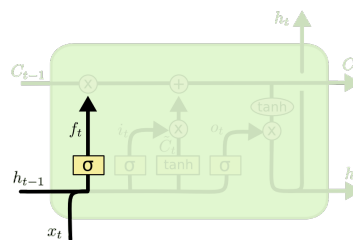
No.	1	2	3	4	5	6	7	8	9	10
Answers	B	ACD	AC	B	BC	AC	D	AB	BD	BC

1. In the McCulloch-Pitts model of an artificial neuron, how is the neuron's output determined? [one answer is correct]
 - a. By integrating over time and applying a sigmoid function
 - b. By summing weighted inputs and comparing to a threshold.
 - c. By calculating average spike rate over all inputs.
 - d. By maximizing input entropy across all synapses.
2. Which of the following are true about Hamming distance and its role in Hopfield Networks? [more than one answer is correct]
 - a. The Hamming distance counts the number of differing bits between two binary vectors.
 - b. Hopfield Networks always converge to the stored pattern with the maximum Hamming distance.
 - c. The network retrieves the pattern with the smallest Hamming distance to the test input.
 - d. Hamming distance can be used to measure similarity between a test input and stored patterns.
3. What conditions must a stored pattern satisfy? [more than one answer is correct]
 - a. The update rule applied to each unit must leave the pattern unchanged.
 - b. The Hamming distance to all other patterns must be zero.
 - c. The sign of the weighted sum of inputs to each neuron must match its state.
 - d. The pattern must be orthogonal to all others in the network.
4. What does the margin represent in the perceptron learning rule? [one answer is correct]
 - a. The maximum distance between any two misclassified points.
 - b. The distance between the separating hyperplane and the closest correctly classified point.
 - c. The number of updates until convergence.
 - d. The angle between the target vector and the current weight vector.
5. In what scenario is a deep learning model generally preferred over a traditional machine learning model? [more than one answer is correct]
 - a. You are building a credit risk model using structured tabular data like income, credit history, and age, with around 2000 samples and a need for model interpretability.
 - b. You are tasked with classifying medical X-ray images into diagnostic categories. You have access to tens of thousands of labeled images, but no handcrafted features are available.
 - c. You are designing a tool to analyze time-series sensor data from wearable devices (e.g., heart rate, motion, temperature) to detect abnormal health events. The dataset consists of millions of continuous readings.
 - d. You are developing a churn prediction model for a telecom provider using billing data, contract length, and customer service history from about 5000 customers.
6. Consider these training strategies in the backpropagation algorithm applied to a neural network on a dataset with 10 000 samples:
 - Strategy 1: the model parameters are updated once after computing gradients on all samples.
 - Strategy 2: the model updates its parameters 10 000 times per epoch, using a single sample at each step.
 - Strategy 3: the dataset is split into small groups of 64 samples, and the model updates after processing each group.

Which of the following statements are correct? [more than one answer is correct]

- a. Strategy 1 performs only one parameter update per epoch.
- b. Strategy 2 ensures maximum stability of gradient estimates.
- c. Strategy 3 is commonly used because it provides a good trade-off between efficiency and convergence speed.
- d. Strategy 1 is typically referred to as stochastic gradient descent.

7. In a convolutional neural network (CNN) with (H_k, W_k) kernel size, (H_{in}, W_{in}) input image size, and (H_p, W_p) padding size, how is the output image size computed? [one answer is correct]
- $(H_{in} - H_k - H_p + 1, W_{in} - W_k - W_p + 1)$
 - $(H_{in} - H_k + H_p - 1, W_{in} - W_k + W_p - 1)$
 - $(H_{in} - H_k + H_p, W_{in} - W_k + W_p)$
 - $(H_{in} - H_k + H_p + 1, W_{in} - W_k + W_p + 1)$
8. Which of the following statements are true about CNNs? [more than one answer is correct]
- CNNs are preferred over multi-layer perceptron for image recognition tasks because they preserve spatial information.
 - Kernels in CNN models are also weights which will be learned via back-propagation process.
 - A multi-layer perceptron is not used in CNNs because they would involve learning too many kernel parameters and do not preserve spatial information.
 - Convolution on a RGB image implies the usage of multiple channels, sharing the same 2D learned filter.
9. At time step t , the LSTM cell in the figure processes the inputs x_t and h_{t-1} . What is the function of the forget gate in this step? [more than one answer is correct]



- It decides how much of the previous hidden state h_{t-1} should be passed to the output.
 - It determines how much of the previous cell state should be kept.
 - It resets the hidden state h_t to zero for the next time step.
 - It multiplies C_{t-1} elementwise with a sigmoid gate output to control memory flow.
10. Which of the following statements are correct about autoencoders [more than one answer is correct]
- The decoder is used to obtain a latent low-dimensional representation of the data.
 - Unlike standard convolution, transposed convolution spreads each input pixel value over a larger area by reversing the process. It can be visualized as inserting zeros between elements in the input and then applying the standard convolution.
 - The latent space can be used to sample and generate new data.
 - Autoencoders are composed by two subnetworks: a generator and a discriminator.

Part II - Open Questions [50 points]

1. (15 points) Answer the following open questions about the Universal Approximation Theorem (UAT).

- a) (4) What does the UAT state?

Any continuous function $f: [0,1]^n \rightarrow [0,1]$ can be approximated with arbitrary precision by a neural network with at least 1 hidden layer with a finite number of weights.

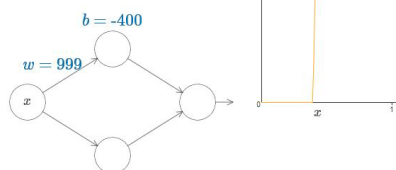
- b) (3) What is the translation of the statement in “mathematical terms”?

Suppose we have a function $f(x)$ that we would like to compute within some desired accuracy $\epsilon > 0$.
It is guaranteed that by using enough hidden neurons we can always find a neural network whose output $g(x)$ satisfies $|g(x) - f(x)| < \epsilon$ for all inputs x .

- c) (8) Considering a function with 1 input and 1 output, describe the 4 steps of the visual proof of the UAT (use figures to support the explanation).

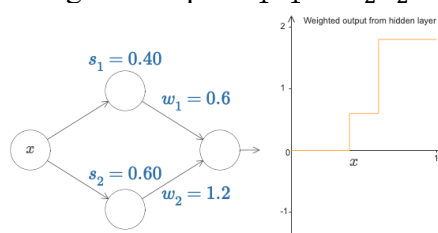
Step 1

By operating on one of the neurons, generate a step function (easier to sum up). We do this by fixing the weight w to be some very large value and then setting the position of the step by modifying the bias. The step is at position $s = -\frac{b}{w}$.



Step 2

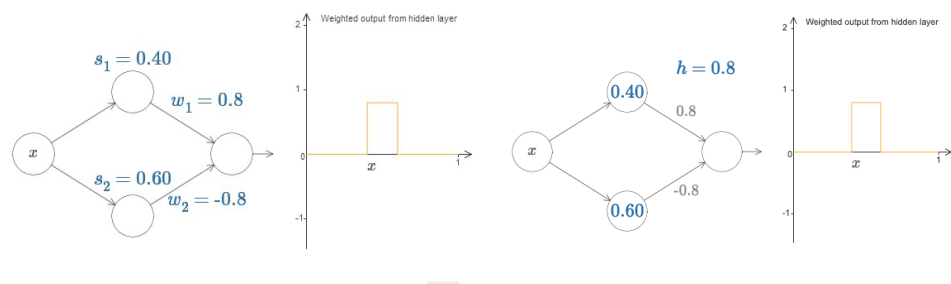
Let's focus on the whole network and suppose the hidden neurons are computing step functions parametrized by the step points s_1 and s_2 . Let's consider the weights w_1 and w_2 on the hidden-to-output connections. We can compute the weighted output $w_1 a_1 + w_2 a_2$ from the hidden layer:



Step 3

Working on the second weight, obtain a “bin” function, starting at point s_1 and ending at s_2 .

As illustrated below, by using the other neuron to make a step function, and setting opposing weights in the second layer, we can effectively approximate a bin and control its position, size and height. We can hence use just one parameter $h = w_1 = -w_2$ to control both the weights!

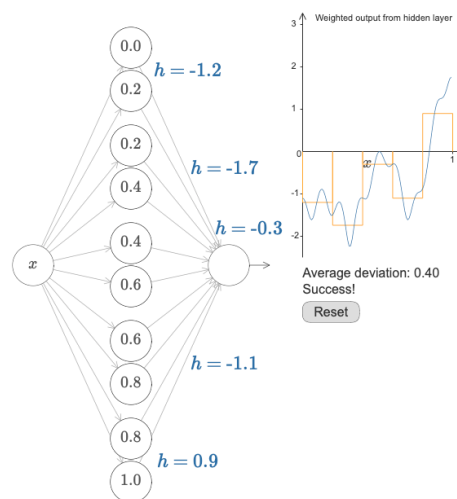


Step 4

By adding pairs of hidden neurons, we can create multiple bins.

More generally, if we divide the interval $[0,1]$ into N subintervals and use N pairs of hidden neurons to set up peaks at different desired heights, we can generate a histogram that discretizes the function.

Playing with the heights and the number of bins, we can approximate a function up to a desired precision.



2. (15 points) Consider a Hopfield network storing a single pattern $\vec{\xi}$.

a. (4) Give the definition of attractor and of the reversed state.

1pt for the mentioning stability (An attractor is a stable state of the Hopfield network where the neuron updates stop changing the state).
1pt for memory retrieval/attraction from another input state (It represents a memory that the network can recall from nearby initial conditions).

The reversed state of a pattern is obtained by flipping all its bits. In some cases, both the attractor and its reversed state can be stable attractors.

b. (6) Prove that $\vec{\xi}$ is an attractor of the network defined by the weights $w_{ij} = \frac{1}{N} \xi_i \xi_j$.

Proof:

Let's consider the total net input $h_i = \sum_{j=1}^N w_{ij} x_j$

$$\Rightarrow h_i = \sum_j \frac{1}{N} \xi_i \xi_j x_j = \frac{1}{N} \xi_i \sum_j \xi_j x_j = \frac{1}{N} \xi_i \left[\sum_{j=\text{correct}} \xi_j x_j + \sum_{j=\text{wrong}} \xi_j x_j \right] = \frac{\xi_i}{N} (N_{\text{correct}} - N_{\text{wrong}})$$

$$\Rightarrow O_i = \text{sgn}(h_i) = \text{sgn} \left[\frac{\xi_i}{N} (N_{\text{correct}} - N_{\text{wrong}}) \right] = \begin{cases} \xi_i & \text{if more than or exactly } \frac{N}{2} \text{ are correct,} \\ -\xi_i & \text{otherwise/opposite case.} \end{cases}$$

$\Rightarrow \xi_i$ is an attractor! AND in this simple case we also have another attractor at the reversed state $-\xi$.

c. (5) Design a network of 3 neurons (i.e., compute the weights) that remembers a pattern $\vec{\xi} = [1, -1, 1]$. Then prove that $\vec{\xi}$ is stable for the designed network.

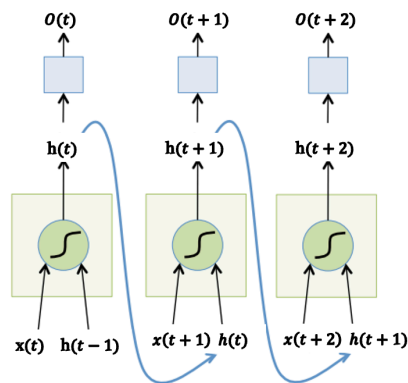
$$W = \frac{1}{N} \vec{\xi} \vec{\xi}^T = \frac{1}{3} \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 \end{pmatrix} = \frac{1}{3} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

Is $\vec{\xi}$ stable?

$$\begin{aligned} \vec{0} = \text{sgn}(W \cdot \vec{\xi}) &= \text{sgn} \left[\frac{1}{3} \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right] = \text{sgn} \left[\frac{1}{3} \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix} \right] \\ &= \text{sgn} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \vec{\xi} \end{aligned}$$

$\Rightarrow$ the answer is YES!

3. **(20 points)** Answer the following questions about Recurrent Neural Networks (RNNs):
- a. **(5)** Provide an explanation of the following figure. Explain what each of the symbols represents, how $h(t)$ is computed, and what is the main characteristic of this network compared to simpler architectures, e.g., a multi-layer perceptron.



The figure shows an unfolded RNN, where the hidden state acts as memory, storing information from previous time steps.

$x(t)$ and $O(t)$ are the input and output at time step t . $h(t)$ is the hidden state at time t , computed as follows:

$$h(t) = \text{sigmoid} \left(W \begin{bmatrix} x(t) \\ h(t-1) \end{bmatrix} \right)$$

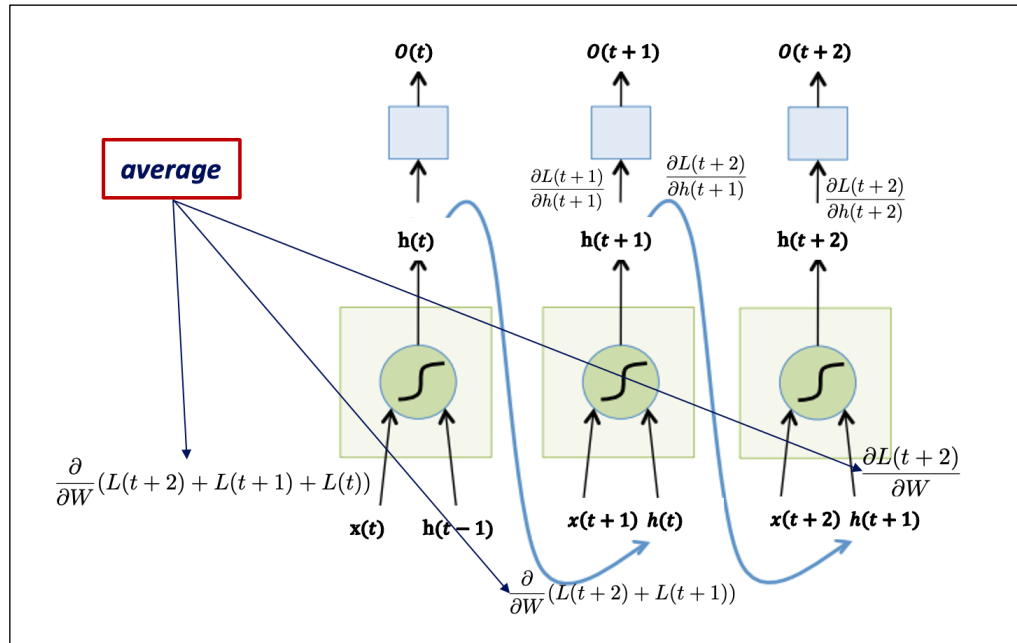
Each new input updates the hidden state based on both the current input and the past memory. This allows the network to process sequences and maintain context over time.

- b. **(5)** What algorithm is used to train an RNN and what are the core principles?

Backpropagation Through Time (BPTT)

- Unfold (copy) the RNN in time during the forward pass
- Treat the unfolded RNN as a multi-layer NN
- The weight updates are computed for each copy in the unfolded network, then averaged

- c. (5) Write the expression for the total gradient of the loss with respect to the shared weights across the 3 time steps.



- d. (5) When training a vanilla Recurrent Neural Network on long sequences, what difficulty can arise in the way gradients behave across time steps? And how does the architecture of an LSTM address the difficulty you described above?

When training vanilla RNNs on long sequences, the gradients can either shrink to zero (vanishing gradients) or grow uncontrollably (exploding gradients).

This happens because the same weight matrix is repeatedly applied during backpropagation. If its singular values are small, gradients vanish; if large, they explode. As a result, the network struggles to learn long-term dependencies.

LSTMs mitigate this by using gated mechanisms (input, forget, and output gates) that regulate the flow of information. The sigmoid function outputs values between 0 and 1, controlling how much information is allowed through. The tanh function keeps values bounded between -1 and 1, helping to prevent exploding gradients. By combining these mechanisms, LSTMs better manage gradient flow and enable learning over longer sequences.